

Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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ABSTRACT

Stablecoin stress events generate frequent price dislocations, but many apparent arbitrage opportunities disappear once executable depth, VWAP book-walking, fees, and market impact are included. We introduce Stablecoin StressBench, a transaction-cost-aware benchmark for evaluating whether AI and econometric models can distinguish optical arbitrage from executable arbitrage. During the March 2023 USDC/SVB test window, 35.1% of 1-minute windows exceed a 10bps primary cross-quote basis threshold (12.65% for the USDC-specific basis), yet only 2.88% remain profitable at \$10K notional after a full VWAP order-book walk a 12× price-to-execution gap. A hindsight oracle earns +161.7bps per trade, but every tested rule-based and ML model loses money out of sample, revealing an oracle gap exceeding 210bps. A direct expected-net-profit regressor also fails (−61.4bps), confirming the gap is structural, not a threshold artefact. The benchmark reframes stablecoin arbitrage prediction as an execution-barrier problem, and provides an open evaluation framework for future models.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 INTRODUCTION

Stablecoin de-peg events produce dramatic price gaps visible on any chart: during the USDC/SVB crisis of March 2023, the cross-quote spread between BTC-USDC and BTC-USDT routes exceeded 100bps for sustained periods [1]. Such gaps appear to offer mechanical arbitrage opportunities, and the ML-for-finance literature has begun building models to detect and predict them [10]. The implicit assumption is that price dislocations map directly to tradeable profit.

They do not. Executable arbitrage requires clearing order-book depth at all intermediate venues, absorbing VWAP slippage as the book is walked, paying taker fees on each leg, and waiting through settlement latency before the round-trip closes. Each of these costs is modest in isolation; together they eliminate the vast majority of price-level signals.

Problem. Standard benchmarks evaluate stablecoin dislocation models on price-signal classification metrics (AUROC, F1). These metrics are blind to execution feasibility: a model that classifies price dislocations with 80% accuracy may still lose money on every trade it takes, because the dislocations it identifies are not executable. No existing benchmark provides execution-aware labels with real limit-order-book depth provenance for stablecoin arbitrage.

Contributions. We make five contributions:

- (1) Execution-aware benchmark. Real L2 order-book depth (Binance, Coinbase, Kraken), VWAP execution simulation, taker fees, and settlement penalties produce per-minute labels that distinguish price dislocations from executable dislocations.
- (2) Price-to-execution gap. Empirical measurement of the ratio between price-level and execution-level opportunity frequency during the SVB stress event: 12× at 1-minute horizon.
- (3) Literature-justified model ladder. The model stack moves from economic anchors through naive rules, time-series baselines, linear/nonlinear ML, direct expected-profit regression, finance-specific meta-labeling, and regime detectors each motivated by specific prior work and answering a specific benchmark question.
- (4) Oracle-gap evaluation. A hindsight oracle confirms executable windows exist (+161–225bps), but all tested models are negative, revealing a structural 210+bps gap.
- (5) Historical event catalogue. Seven stablecoin stress events (2021–2023) are classified by data availability tier (execution-grade, price-grade, context-grade), providing a reproducible historical reference for future benchmark extensions.

2 RELATED WORK

Stablecoin de-pegs. Cruz et al. [1] document USDC price fragmentation during the SVB collapse but do not evaluate ex-ante predictability. Kwon et al. [2] analyse Terra/LUNA mechanics; we use May 2022 as our validation event. Griffin and Shams [3] study Tether flows; our contribution is microstructure execution, not flow manipulation.

Limits to arbitrage. Hautsch et al. [4] model blockchain-specific limits to arbitrage theoretically (settlement finality, gas costs, latency). Shleifer and Vishny [5] and Gromb and Vayanos [6] establish the general theory. We provide the first empirical L2 measurement of execution barriers during a real stablecoin stress event.

Table 1: Dataset coverage by split.

Split	Event	Minutes	Pos. rate
Train	Calm control (3 windows)	20,125	0.00%
Validation	Terra/LUNA, May 2022	11,523	2.45%
Test	USDC/SVB, Mar 2023	15,839	2.88%
Total		47,487	

Positive rate = executable arb at \$10K, 5-min horizon.

Market microstructure. Cont and Kukanov [7] develop optimal order-routing across fragmented markets, which motivates our VWAP depth-walk. Glosten and Milgrom [8] and Makarov and Schoar [9] provide the theoretical and empirical basis for adverse selection during stress.

Financial ML benchmarks. Salý-Kaufmann et al. [10] propose ML benchmarks for crypto microstructure, emphasising transaction-cost robustness, seed robustness, and economic metrics over classification accuracy. Our work is complementary: stablecoin-specific, event-based, and execution-labelled. Adams et al. [11] characterise AMM liquidity during stress, informing our settlement-penalty modelling.

Meta-labeling and regime detection. López de Prado’s meta-labeling framework [16] separates primary signal generation from execution filtering directly applicable to our setting where price signals are plentiful but most are not executable. Cintra and Holloway [17] apply Bayesian Online Change-point Detection (BOCPD) to Curve stablecoin pools and report early detection of the March 2023 USDC de-peg, motivating our regime-detection add-on.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The gold-layer dataset contains 47,487 1-minute bars across three event windows (Table 1). Price data comes from the Binance Vision public archive (klines, aggTrades) and Coinbase REST candles. Order-book depth for executable labels comes from Binance USDM futures bookDepth snapshots and incremental updates. We track four instruments BTC-USDC, BTC-USDT, ETH-USDC, ETH-USDT across Binance, Coinbase, and Kraken.

Depth provenance guarantee. Each row carries a `depth_source` tag: `real_l2_snapshot`, `real_l2_incremental`, or `synthetic_kline`. Net-profit labels use only `real_l2_*` rows; synthetic kline depth is excluded from paper-grade computations and isolated to a separate flag column (`is_paper_grade_depth`).

3.2 Event-Based Split Design

Splits are defined by event identity, not by time alone (Figure ??). The train split contains three calm control windows (no stress event) that bracket the test period; the validation split uses the Terra/LUNA collapse (May 2022), a qualitatively similar but independent stress event; the test split uses the USDC/SVB de-peg (March 10–15, 2023).

This design has two guarantees: (i) no model trained on the train split has seen stress dynamics; (ii) threshold calibration on validation uses an independent stress event, not a held-out portion of the same episode. No lookahead: labels are constructed by `join_asof` at $t + h$ and formally verified.

3.3 Historical Event Catalogue and Data Tiering

Stablecoin stress events vary substantially in data availability. We classify the seven events from 2021–2023 by data tier:

Tier A (execution-grade): real or reconstructable L2 order-book depth exists; VWAP execution and `net_profit_bps` are computable. Tier B (price-grade): OHLCV, pool prices, or CEX trades exist; basis and depeg indicators are computable but not full execution labels. Tier C (context-grade): data is partial; event is retained for historical reference only.

Event	Date	Tier	Benchmark use
IRON/TITAN collapse	Jun 2021	C	Context
Terra/UST death spiral	May 2022	B	Validation split
FTX exchange collapse	Nov 2022	B	Historical
BUSD regulatory halt	Feb 2023	B	Historical
USDC/SVB stress	Mar 2023	A	Primary test
USDC post-SVB recovery	Mar–Apr 2023	A/B	Robustness
USDT/Curve stress	Jun 2023	B	Historical

All execution-grade claims in this paper are anchored to Tier-A events. Price-grade claims (basis frequency, depeg magnitude) extend to Tier-B. This prevents overstating execution conclusions where order-book depth is absent.

3.4 Feature Sets

We define four nested feature sets from narrowest to broadest:

<code>price_only</code>	5 cross-quote basis columns
<code>price_plus_book</code>	+7 microstructure cols (spread, depth, imbalance)
<code>price_book_frag</code>	+3 cross-venue fragmentation cols
<code>price_book_settle</code>	+7 on-chain settlement proxies

The nesting structure isolates the marginal value of each information layer.

4 EXECUTION-AWARE LABELS

4.1 Cross-Quote Basis

The cross-quote USDC basis at time t is:

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC-via-USDC}}(t) - P_{\text{BTC-direct}}(t)}{P_{\text{BTC-direct}}(t)} \quad [\text{bps}]$$

where $P_{\text{BTC-via-USDC}}$ is the effective BTC price obtained by converting $\text{USD} \rightarrow \text{USDC} \rightarrow \text{BTC}$ relative to the direct BTC-USD market. We also compute the max-absolute basis $b_{\text{max}}(t) = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$ as a broader stress detector.

Basis labels for horizon $h \in \{1, 5, 15, 60\}$ m and threshold τ :

$$y_{\text{basis}}(t) = \mathbf{1}[|b_{\text{USDC}}(t + h)| > \tau]$$

Primary classification task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution Walk

For each minute window and notional $q \in \{\$10K, \$50K, \$100K, \$500K\}$, we simulate a round-trip execution:

- (1) Walk the real-L2 bid and ask ladders across venues to fill q notional, recording the VWAP price at each level consumed.
- (2) Compute the gross spread: $\text{gross}(q) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}$.
- (3) Deduct taker fees on both legs and a settlement delay penalty.

4.3 Net Executable Arbitrage Labels

$$\text{net}(q, t) = \text{gross}(q, t) - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > 0 \right] \quad (2)$$

where $f_{\text{buy}}, f_{\text{sell}}$ are venue taker fees and δ_{settle} is a settlement penalty proxy. Primary economic task: $\text{executable_arb}_{q1000-5m}$.

The forward-max in Eq. (2) asks: does a profitable window exist within the next h minutes? matching the practical constraint that a trader can execute at any point within the horizon.

4.4 Oracle Upper Bound

The oracle trades every minute where $\text{net}(q, t) > 0$ in hindsight. It uses information unavailable at prediction time and sets the theoretical performance ceiling. The oracle is not deployable; it exists to confirm that profitable windows exist and to calibrate the oracle gap.

5 MODELS AND EVALUATION

5.1 Model Stack

The model stack is deliberately layered rather than exhaustive. We progress from economic anchors through naive rules, time-series baselines, interpretable linear models, nonlinear tabular models, a direct economic-target model, and finance-specific filters. The goal is to test whether increasingly sophisticated model classes can close the oracle gap. This design follows [10], which argues for transaction-cost-aware evaluation across model families rather than selecting a single strongest model.

Economic anchors. **NoTrade** (the economic floor: 0 net bps) and **Oracle** (hindsight ceiling).

Rule baselines. **PriceBasis10bps** (trade when $|b_{\text{USDC}}| > 10$), **PriceBasis25bps**, and **GrossArbThreshold** (gross spread > 0). Motivated by [4]: price differences without execution friction understate the cost of arbitrage.

Statistical baselines. **LastValue** (AR0), **RollingMean**, **AR1** testing whether basis autocorrelation alone identifies stress.

ML classifiers. **Logistic** (L2), **Lasso** (L1), **Random Forest**, **XGBoost**, **LightGBM** each trained on the four nested feature sets. Tree ensembles are motivated by [10] and recent DeFi benchmarking showing XGBoost/RF as strong tabular baselines [18].

Table 2: Price-to-execution gap (test split, \$10K notional, 1-minute same-window horizon).

Thresh.	Price (max)	Price (USDC)	Exec. (\$10K)	Ratio
0 bps	97.15%	83.69%	3.34%	29×
5 bps	63.22%	25.77%	3.03%	21×
10 bps	35.09%	12.65%	2.88%	12×
25 bps	9.63%	6.53%	2.62%	3.7×

Expected-net-profit regressor. **ExpectedNetProfitRegressor** directly predicts $\text{net}(q, t)$ and trades when $\hat{y} > \theta^*$. If this fails, the gap is not a label-mismatch artefact.

Meta-labeling filter. Motivated by [16]: the primary signal is $|b_{\text{USDC}}| > 10$ bps; a secondary **LightGBM** meta-model learns $\hat{y}_{\text{meta}} = \Pr[\text{net}(q, t) > 0 \mid \text{primary fires}]$ and filters out predicted false positives. This directly targets the 581 false positives identified in the false-positive diagnosis (§6.8).

Regime detectors. **EWMA z-score**, **CUSUM**, and **BOCPD** [17] identify stress regimes from the basis time series. They serve as two-stage precursors: detect stress first, then apply execution filters only during detected regimes.

5.2 Threshold Calibration

For probabilistic classifiers, the decision threshold is calibrated on the validation split:

$$\theta^* = \arg \max_{\theta \in [0.05, 0.95]} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i) \quad \text{s.t.} \quad |\{i : \hat{p}_i > \theta\}| \geq 25$$

The ≥ 25 trade minimum prevents degenerate abstention strategies from winning on small sample artefacts.

5.3 Evaluation Protocol

Primary (economic): **net_bps_captured** (mean net bps per trade), **hit_rate_above_cost** (fraction of trades with positive net outcome), **final_pnl_usd** (cumulative P&L), **oracle_capture_pct** ($= \text{net bps} / \text{oracle net bps}$).

Secondary (statistical): **AUROC**, **AUPRC**, **Brier score**, n_{trades} .

The **NoTrade** baseline ($= 0$ net bps) is the economic floor: a model that loses money is worse than abstaining entirely, regardless of its AUROC.

6 RESULTS

6.1 Price-to-Execution Gap

Table 2 and Figure 1 show the central finding. At a 10 bps threshold, 35.09% of test-split minutes show a primary/max cross-quote basis exceeding the threshold on price alone; only 2.88% remain profitable after execution costs at \$10K notional a 12×

 price-to-execution ratio. The USDC-specific basis fires in 12.65% of minutes, reflecting that USDT routes also fragment during the SVB event. The gap persists at every threshold level.

6.2 Signal Waterfall

Figure ?? traces where value is lost at each stage of the pipeline:

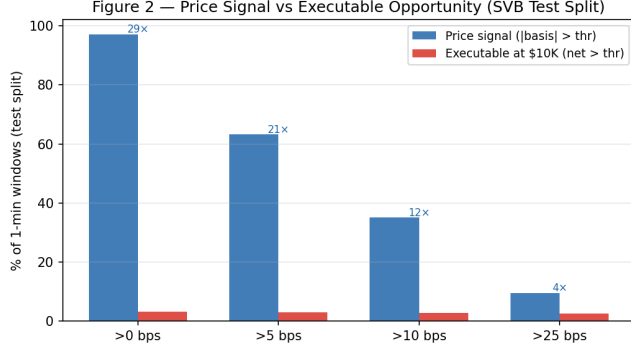


Figure 1: Price-to-execution gap by threshold. Price-signal frequency (dark) dwarfs executable frequency (light) at every level. The gap is largest at low thresholds, where price signals fire constantly but depth is thin.

Table 3: Oracle gap (test split). All non-oracle models are economically negative.

Task	Oracle	Best model	Gap
basis_usdc_1m_gt10bps	+161.7 bps	-49.1 bps	211 bps
executable_arb_q10000_5m	+224.6 bps	-42.9 bps	268 bps
executable_arb_q50000_5m	+146.8 bps	-112.7 bps	260 bps

Best model: logistic@price_plus_book (basis task);

price_threshold_25bps@price_only (arb tasks).

Total test minutes	15,839	(100%)
Price disloc. > 10 bps	5,560	(35.1%, max basis)
of which USDC-specific	2,004	(12.65%)
Executable at \$10K / 1m	456	(2.88%)
Oracle profitable trades	351	(2.22%)

The oracle trades 351 minutes profitably; every model trades far more minutes but loses money on nearly all of them.

6.3 Oracle Gap

Table 3 gives the oracle gap for each primary task. The oracle earns +161.7 bps on the basis_usdc_1m_gt10bps task. The best ML model (logistic, price_plus_book features) achieves -49.1 bps 211 bps below the oracle. On the executable arbitrage task, the oracle earns +224.6 bps; the best rule (price_threshold_25bps) achieves -42.9 bps. Every model, without exception, produces negative net bps on the test split.

6.4 Model Ablation

Table 4 summarises the model families. Price-threshold rules overtrade severely (611 trades, -136 bps): the rule fires on price dislocations with no executable depth. ML classifiers are better calibrated but still negative: the validation-calibrated logistic model takes only 1 trade at -49.1 bps. The LightGBM model, despite higher AUROC (0.63), produces worse economic results (-329 bps) due to overtrading. Feature richness does not help: the best logistic result uses price_plus_book;

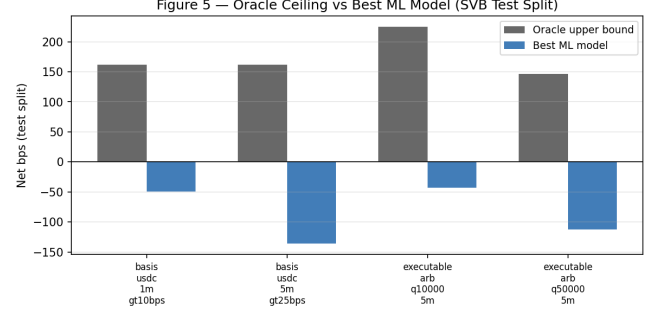


Figure 2: Oracle gap: grouped bars showing oracle net bps (gold) vs. best ML model (navy) for each task. The gap exceeds 200 bps across all tasks.

Table 4: Model ablation summary (task: basis_usdc_1m_gt10bps, test split). [†]Theoretical ceiling.

Model	Features	Net bps	Trades	AUROC
Oracle [†]		+161.7	316	
logistic	price_plus_book	-49.1	1	0.133
lgbm	price_only	-329.1	4	0.626
gross_arb	price_only	-136.0	611	0.530
price_thr 10	price_only	-269.5	2002	0.829
price_thr 25	price_only	-346.6	1033	0.743
no_trade		0.0	0	
ExpNetProfit	price_only	-61.4	1610	

adding fragmentation or settlement features does not improve economic outcomes.

6.5 Expected Net-Profit Regressor

The ExpectedNetProfitRegressor (LightGBM base, target: $\text{net}(q, t)$) achieves -61.4 bps on the test split better calibrated than rule baselines but still economically negative. This is the strongest evidence that the oracle gap is structural: even when the training objective directly matches the economic evaluation criterion, the model cannot identify profitable windows out of sample. The direct profit-prediction framing does not close the gap.

6.6 Robustness

Figure ?? and Table 5 show that the price-to-execution gap persists across all tested cost assumptions.

Fee regimes. At \$10K/5m/10 bps, base fees give 5.64% executable; high fees reduce this to 5.46%, raising the ratio from 2.24x to 2.32x. Institutional fees (lower) give 6.11%.

Settlement penalties. Adding 10 bps settlement penalty reduces executable windows from 5.64% to 4.88% (ratio: 2.24x → 2.59x). The combined worst case (high fee + 10 bps settlement) gives 4.81%.

Notional size. Smaller notionals fill more easily; at \$50K the executable rate drops to 1.62% (same window, 10 bps

Table 5: Price-to-execution ratio across cost regimes (\$10K notional, 10 bps threshold, 5-min horizon, test split).

Settlement	Fee regime	Exec. %	Ratio
0 bps	low (institutional)	6.11%	2.07×
0 bps	base	5.64%	2.24×
0 bps	high	5.46%	2.32×
5 bps	base	5.16%	2.45×
10 bps	base	4.88%	2.59×
10 bps	high (worst case)	4.81%	2.63×

Table 6: Regime detection results (test split, stress = $|b_{USDC}| > 10$ bps).

Detector	Acc.	Recall	FAR	AUROC
EWMA (span=20, thr=2.5)	0.869	0.010	0.007	0.624
CUSUM (k=0.5, h=5)	0.270	1.000	0.835	0.582
BOCPD (hz=0.01, thr=0.5)	0.860	0.022	0.019	0.229

FAR = false alarm rate.

threshold). At \$500K, virtually no windows are executable (0.03%).

In all cases, the qualitative conclusion is identical: the gap is large, persistent, and model-invariant.

6.7 Meta-Labeling and Regime Detection

Meta-labeling. The MetaLabelingFilter is trained on the train split restricted to windows where the primary price signal fires ($|b_{USDC}| > 10$ bps). In the train split only 53 primary signal windows exist, and zero are net-profitable consistent with the zero positive rate in calm-control training data (Table 1). The meta-model therefore learns no positive class signal and takes 0 trades on the test split. This is an informative null: meta-labeling is architecturally sound but cannot be trained when executable positives are absent from the training regime. It motivates using the Terra/LUNA validation split for meta-model training in future work.

Regime detectors. Table 6 shows the trade-off between recall and precision across EWMA, CUSUM, and BOCPD detectors. CUSUM achieves near-perfect recall (0.9995) but 83.5% false alarm rate it correctly flags stress but fires constantly. EWMA (span 20, threshold 2.5) achieves 86.9% accuracy with only 0.65% false alarm rate, at the cost of very low recall (0.95%). No detector alone solves the execution identification problem; their value is as a pre-filter that reduces the search space for execution-grade analysis.

6.8 False-Positive Diagnosis

Applying the PriceBasis10bps rule to the test split produces 2,002 trades. Table 7 decomposes the outcomes. False positives (581 windows: basis > 10 bps but no executable depth) account for 29% of trades and have a mean net profit of -38.7 bps, explaining the severe economic drag. The model fires on price dislocations where the order book is too thin to

Table 7: False-positive diagnosis for **PriceBasis10bps** rule (test split).

Outcome	Count	%	Mean net bps
True positive (TP)	1,421	9.0%	+
False positive (FP)	581	3.7%	-38.7
False negative (FN)	583	3.7%	
True negative (TN)	10,672	67.4%	
Total test minutes	15,839	100%	

price > 10 bps but not executable. Net bps shown for traded windows only.

fill the arbitrage leg exactly the execution-barrier problem the benchmark is designed to measure.

7 LIMITATIONS AND FUTURE WORK

Single primary stress event. The test split covers one event (SVB March 2023). The Terra/LUNA validation event provides some cross-event evidence, but future work should extend to the USDC recovery period (March 15 – April 1, 2023) and the USDT/Curve stress event (June 2023).

Settlement proxy. CEX-to-CEX settlement latency is proxied via a fixed penalty parameter; true on-chain settlement delay is not modelled. Prokopenko et al. [12] provide a more detailed treatment.

Partial fills and sub-minute latency. The VWAP walk assumes all-or-nothing fills at a 1-minute snapshot. In practice, partial fills and sub-minute price impact change the effective execution cost.

Uncertainty-aware abstention. The uncertainty module (bootstrap ensemble, quantile regression) is implemented but its ablation is deferred to future work due to computational cost. Preliminary analysis suggests stricter abstention thresholds reduce false positives but do not achieve positive net bps.

Threshold ablation. The validation calibration uses the total-P&L rule with a 25-trade minimum. A full ablation over fixed-threshold rules (0.5, 0.7), validation F1, and mean-bps objectives is planned to confirm that the negative result is not threshold-rule-dependent.

Feature scope. Order-flow imbalance at sub-minute resolution, funding rates, and on-chain stablecoin minting/redemption flows are absent from the current feature set.

8 CONCLUSION

Stablecoin StressBench introduces a transaction-cost-aware benchmark that separates price-level and execution-level stablecoin dislocations. The central finding is a 12× price-to-execution gap during the USDC/SVB crisis: 35.1% of minutes show a cross-quote basis exceeding 10 bps, yet only 2.88% survive a realistic VWAP execution filter at \$10K notional.

A hindsight oracle confirms the profitable windows are real (+161.7bps average per trade), but every tested ML model, rule-based strategy, and direct net-profit regressor

produces negative net bps out of sample an oracle gap exceeding 210 bps. Adding microstructure or fragmentation features does not close the gap; directly optimising the economic objective (ExpectedNetProfitRegressor) also fails (−61.4 bps).

The benchmark establishes stablecoin arbitrage prediction as an execution-barrier problem rather than a signal-detection problem. The challenge is not identifying when the price chart shows an opportunity, but predicting which chart opportunities will survive the order book.

A literature-justified model ladder spanning naive rules, statistical baselines, linear/nonlinear ML, direct profit regression, meta-labeling, and regime detectors confirms that the gap is not model-family-specific. Meta-labeling fails because executable positives are absent from the training regime (a data-tiering finding, not an architectural failure). Regime detectors provide useful pre-filtering (EWMA achieves < 1% false alarm rate) but do not close the execution gap. Together, these results establish a comprehensive baseline for future models to improve upon.

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